

String

```
1.import java.io.*;
import java.util.*;
```

```
public class Solution {

    public static void main(String[] args) {
        Scanner s=new Scanner(System.in);
        String st = s.next();
        char [] ch = st.toCharArray();
        String rev = "";
        for(int i=ch.length-1;i>=0;i--){
            rev+=ch[i];
        }
        System.out.println(rev);
    }
}
```

```
2. import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();           // read the string
        String output = input.replace(" ", ""); // remove all spaces
        System.out.println(output);             // print without spaces
    }
}
```

```
3.import java.io.*;
import java.util.*;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        // remove vowels using regex
        String output = input.replaceAll("[AEIOUaeiou]", "");

        System.out.println(output);
    }
}
```

```
4.import java.io.*;
import java.util.*;
```

```
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str1 = sc.nextLine(); // first string
        String str2 = sc.nextLine(); // second string

        String result = str1 + str2; // concatenation
        System.out.println(result); // print result
    }
}
```

```
5.import java.io.*;
import java.util.*;
```

```
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        int count = 0;
        for (int i = 0; i < input.length(); i++) {
            char ch = input.charAt(i);
            if ("AEIOUaeiou".indexOf(ch) != -1) { // check if ch is a vowel
                count++;
            }
        }

        System.out.println(count);
    }
}
```

```
6.import java.io.*;
import java.util.*;
```

```
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        // LinkedHashMap preserves the order of insertion
    }
}
```

```

Map<Character, Integer> map = new LinkedHashMap<>();

for (char ch : input.toCharArray()) {
    map.put(ch, map.getOrDefault(ch, 0) + 1);
}

for (Map.Entry<Character, Integer> entry : map.entrySet()) {
    System.out.println(entry.getKey() + " " + entry.getValue());
}
}
}

```

```

7.import java.io.*;
import java.util.*;

```

```

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();

        int vowels = 0, consonants = 0, digits = 0, special = 0;

        for (int i = 0; i < input.length(); i++) {
            char ch = input.charAt(i);

            if (Character.isLetter(ch)) {
                // check vowels
                if ("AEIOUaeiou".indexOf(ch) != -1) {
                    vowels++;
                } else {
                    consonants++;
                }
            } else if (Character.isDigit(ch)) {
                digits++;
            } else {
                special++;
            }
        }

        System.out.println("vowels:" + vowels);
        System.out.println("consonants:" + consonants);
        System.out.println("digits:" + digits);
        System.out.println("special characters:" + special);
    }
}

```

```
}
```

```
8.import java.io.*;
```

```
import java.util.*;
```

```
public class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String input = sc.nextLine();
```

```
        boolean onlyDigits = true;
```

```
        for (int i = 0; i < input.length(); i++) {
```

```
            if (!Character.isDigit(input.charAt(i))) {
```

```
                onlyDigits = false;
```

```
                break;
```

```
            }
```

```
        }
```

```
        if (onlyDigits && input.length() > 0) {
```

```
            System.out.println("only digits");
```

```
        } else {
```

```
            System.out.println("no");
```

```
        }
```

```
    }
```

```
}
```

```
9.import java.io.*;
```

```
import java.util.*;
```

```
public class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String str1 = sc.nextLine();
```

```
        String str2 = sc.nextLine();
```

```
        // Convert to char arrays
```

```
        char[] arr1 = str1.toCharArray();
```

```
        char[] arr2 = str2.toCharArray();
```

```
        // Sort both arrays
```

```
        Arrays.sort(arr1);
```

```
        Arrays.sort(arr2);
```

```

        // Compare sorted arrays
        if (Arrays.equals(arr1, arr2)) {
            System.out.println("The given strings are an anagram");
        } else {
            System.out.println("The given strings are not an anagram");
        }
    }
}

```

```

10.import java.io.*;
import java.util.*;

```

```

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();

        if (s.length() < 2) {
            System.out.println("No");
            return;
        }

        char first = s.charAt(0);
        char second = s.charAt(1);

        if (first == second) {
            System.out.println("No");
            return;
        }

        boolean isAlternating = true;
        for (int i = 0; i < s.length(); i++) {
            if (i % 2 == 0 && s.charAt(i) != first) {
                isAlternating = false;
                break;
            } else if (i % 2 == 1 && s.charAt(i) != second) {
                isAlternating = false;
                break;
            }
        }

        if (isAlternating) {
            System.out.println("Yes");
        }
    }
}

```

```

    } else {
        System.out.println("No");
    }
}
}

```

```

11.import java.io.*;
import java.util.*;

```

```

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();

        if (num >= 1 && num <= 50) {
            System.out.println("The first " + num + " Natural Numbers are:");
            for (int i = 1; i <= num; i++) {
                System.out.print(i + " ");
            }
            System.out.println(); // move to next line after printing numbers
        } else {
            System.out.println("Enter a Valid Input!!!!!!!!!!!!!!!!!!!!!!!!!!!!");
        }
    }
}

```

```

12.import java.io.*;
import java.util.*;

```

```

public class Solution {

    // Recursive method to calculate sum from 1 to n
    public static int sum(int n) {
        if (n == 1) {
            return 1;
        }
        return n + sum(n - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if (n >= 1 && n <= 50) {

```

```

        int total = sum(n);
        System.out.printf("The sum of numbers from 1 to %d : %04d\n", n, total);
    } else {
        System.out.println("Enter a Valid Input!!!!!!!!!!!!!!!!!!!!");
    }
}
}

```

```

13.import java.io.*;
import java.util.*;

```

```

public class Solution {

    // Recursive method to calculate sum of digits
    public static int sumOfDigits(int n) {
        if (n == 0) {
            return 0;
        }
        return (n % 10) + sumOfDigits(n / 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();

        if (num >= 0 && num < 1000) {
            int total = sumOfDigits(num);
            System.out.printf("Sum of Digit:%04d\n", total);
        } else {
            System.out.println("Enter a Valid Input!!!!!!!!!!!!");
        }
    }
}

```

```

14.import java.io.*;
import java.util.*;

```

```

public class Solution {

    // Recursive method to count digits
    public static int countDigits(int n) {
        if (n == 0) {
            return 0;
        }
    }
}

```

```

        return 1 + countDigits(n / 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();

        // Handle the case when input is 0
        if (num == 0) {
            System.out.println("The Count the digits: 1");
        } else {
            int count = countDigits(Math.abs(num)); // in case of negative numbers
            System.out.println("The Count the digits: " + count);
        }
    }
}

```

```

15.import java.io.*;
import java.util.*;

```

```

public class Solution {

    // Static method to check Armstrong number
    public static boolean isArmstrong(int n) {
        int original = n;
        int sum = 0;
        int digits = String.valueOf(n).length();

        while (n > 0) {
            int digit = n % 10;
            sum += Math.pow(digit, digits);
            n /= 10;
        }

        return sum == original;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();

        if (num < 0 || num >= 100000) {
            System.out.println("Invalid Input");
        } else {

```



```
        if (isArmstrong(num)) {  
            System.out.println("yes");  
        } else {  
            System.out.println("no");  
        }  
    }  
}
```